

Predicting Employee Attrition

Business Problem:

Attrition in any company can be a major hurdle. Not only is there time investment being lost, but significant monetary costs become a burden to a company with high turnover. Coordinating with Human Resources and obtaining the data necessary to predict employee turnover could significantly reduce a company's annual losses. Focusing on finding the factors that lead an employee to look elsewhere for employment can allow the company to fix the necessary issues and increase employee retention and morale.

Background:

Today's work culture has shifted away from the times of old when an employee would have strong loyalty to a company and would look to stay with the same company until retirement. According to the Bureau of Labor Statistics, the average time a young employee (25 to 34 years) would stay with a company was only 2.8 years. In my area of work, it takes at least 1 year to fully train someone how to perform all aspects of the job effectively. This means that a young new hire may only contribute about 2 years of work without cost (either monetary or time investment) to the department. According to a Forbes article, replacing an employee can cost at least 33% of their annual salary. In a company with a high turnover, this cost can make up a large portion of their annual losses.

It is unclear whether companies or workers caused this relationship where the average worker feels they need to continuously "job hop". However, as a progressive company, finding ways to retain employees should be a top priority. One way to achieve this is to coordinate with Human Resources and leverage employee data in order to analyze/predict which employees are going to be leaving and why.

These developed models will be instrumental in employee retention and in boosting overall morale in the company.

Data Explanation:

The data for this project was obtained from a public data directory, Kaggle.com. The data is a compilation of different factors that are job related. Many of these variables are on a satisfactory scale. This means that the data is heavily reliant on accurate reporting from the survey. Some columns had to be dropped due to lack of importance in the final modeling process. These columns were “EmployeeCount”, “EmployeeNumber”, “Over18”, and “StandardHours”. No feature engineering was able to be performed with this dataset.

Methods:

Data cleaning and analysis was all performed through Jupyter Notebooks using various Python libraries. The data was first cleaned and assessed for any null values. Exploratory Data Analysis was then performed in order to assess any relationships between the target variable and the various independent variables.

Analysis:

Early analysis shows that age, monthly income, years with the company, and total working years have a correlation with attrition. Figures 1-4 visualize this probable correlation between these variables and attrition. If an employee is young and early in their career with a starting wage, they are more likely to be in the cohort that leaves the company. The correlation matrix in Figure 5 did not aid in visualizing this relationship.

It was decided that three models would be sufficient to run on a first pass through this data. The selected models include Logistic Regression, Random Forest Classifier, and Support Vector Machine. Figures 6-9 detail the results of these models. All of the models performed similarly with respect to the

Accuracy, Precision, Recall, and F1-Score values. These scores indicate that the models were returning the true values and there were very few false positives/negatives. Unfortunately, the models struggled with returning false negative values. This means the company may miss some key employees to attrition that the models missed. Feature importance was conducted using the Random Forest model. The results confirm the initial analysis that young workers who are early in their career are most prone to attrition. It also suggests that work distance from home and overtime amount are deciding factors in attrition. Because the initial results for the models were so close, a separate value was needed in order to determine the best model for the data. The Receiver Operating Characteristic Area Under the Curve (ROC AUC) Score is a value that can summarize multiple factors about model performance into one value. The value ranges from 0 to 1, with 0.5 being the value associated with making random selections. Figure 10 shows the ROC AUC Score output for all the models. The Logistic Regression model performed the best based on these scores. However, the values are all close to 0.5, which means that the models are barely better than guessing which employees will fall into the attrition category.

Conclusions:

The result outputs discussed in the analysis did not aid in determining a model of best fit. All the models can make determinations but are not much better than guessing. On re-analysis of the data, steps will have to be taken to account for the uneven distribution of the data between attrition and no attrition. It is unlikely that the models had enough data to work with. Conversely, the initial analysis in combination with the Random Forest Model provides good insight into what may be causing the attrition in the company's employees. Monthly income (i.e. salary) is a major factor in the employee's decision to leave the company. This factor is something that is fixable. Providing competitive compensation packages will increase the company's employee retention.

Assumptions:

The main assumption in this project is that all the variables are independent of each other. It is also assumed that the data is collected without bias and that the respondents gave the true values during their assessment.

Limitations:

As always, time limitations were a factor.

Challenges:

Small data size was a challenge. It would be beneficial to obtain real world data in order to test these developed models. It is difficult to tell if generated data will allow the models to hold up in every scenario.

Future Uses:

Modeling this type of data will greatly benefit many companies/departments. Using machine learning as a tool to help improve employee retention will be a great leap forward in retaining quality employees. Let this serve as a framework for future ideas and uses.

Recommendations:

In the long term, more data would be useful in developing a model that can predict employee attrition. In the short term, developing a more competitive salary package will increase employee retention.

Implementation Plan:

There are no implementation plans as of yet. More data is needed in order to develop a model that is significantly better than randomly guessing which employees will suffer attrition.

Ethical Assessment:

Analyzing people/employees can be tricky. If a company goes too far in analyzing/recording their employees, the employees may distance themselves from their work and become disenfranchised.

Therefore, a balance must be found between analysis and privacy. Data should only be collected where necessary and there should be transparency about how the data will be used. A conscious effort should be made to avoid becoming “Big Brother” in your employees eyes (Kumar 2023).

Reference List:

Employee tenure in 2022. (n.d.). <https://www.bls.gov/news.release/pdf/tenure.pdf>

Hall, J. (2022, October 12). *The cost of turnover can kill your business and make things less fun*. Forbes. <https://www.forbes.com/sites/johnhall/2019/05/09/the-cost-of-turnover-can-kill-your-business-and-make-things-less-fun/>

Kumar, T. (2023, September 27). *People analytics: Ethical considerations*. AIHR. <https://www.aihr.com/blog/people-analytics-ethical-considerations/>

Appendix:

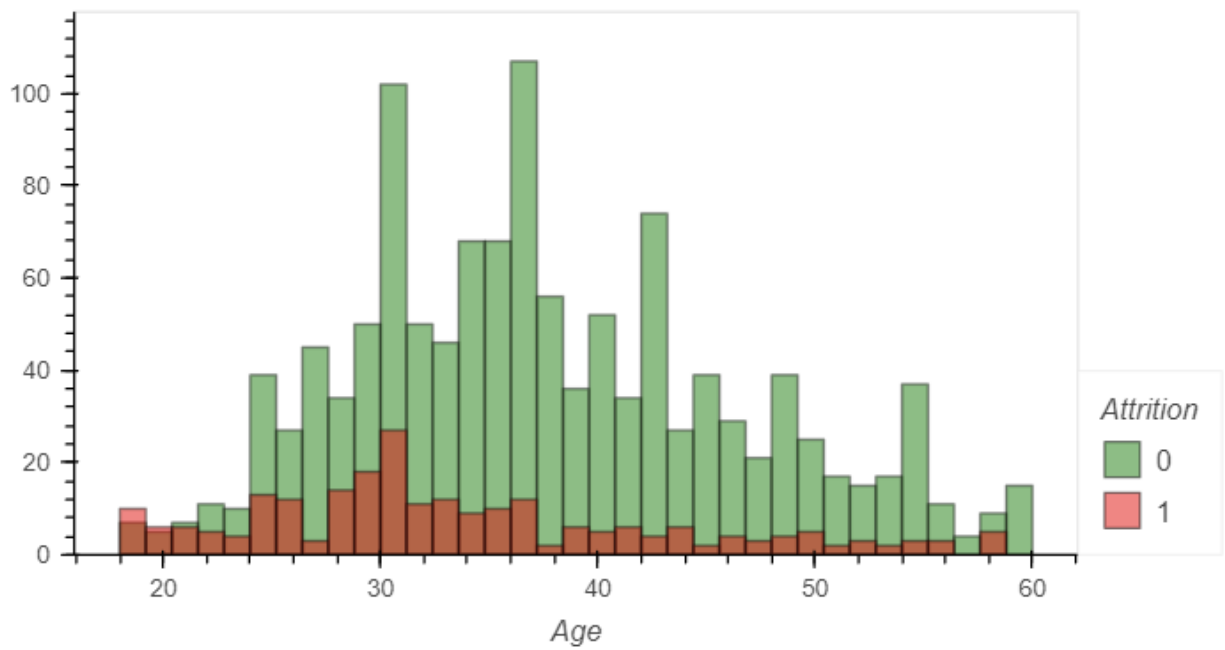


Figure 1 - Age Demographics with Attrition Distribution

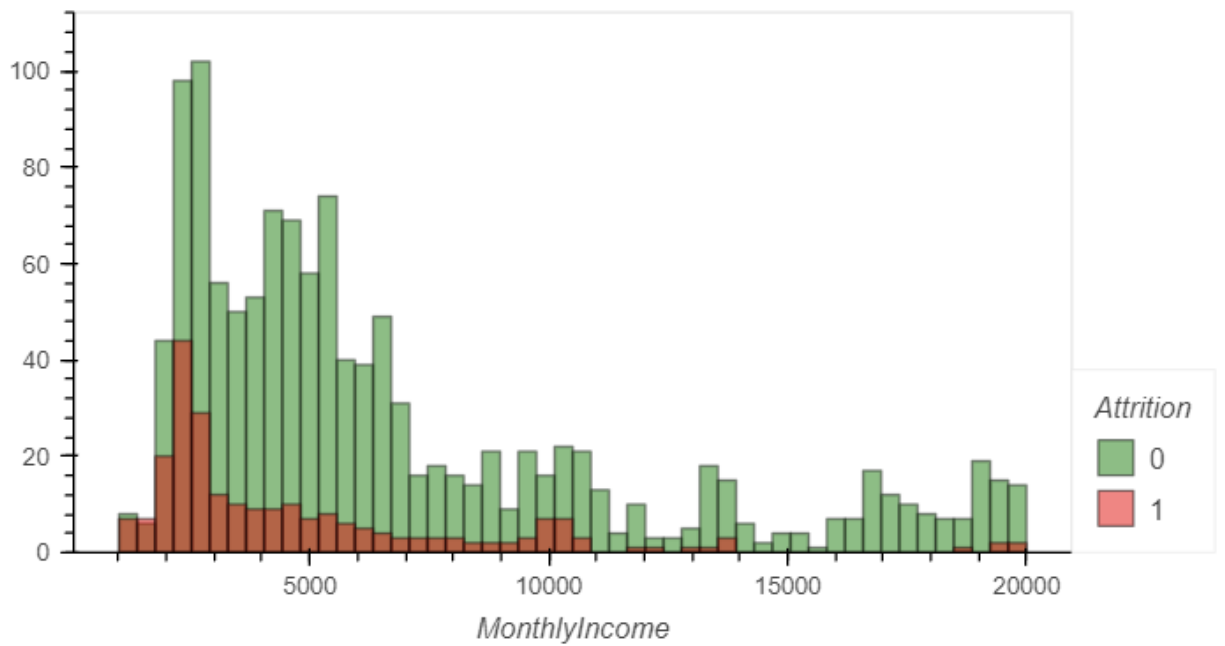


Figure 2 - Monthly Income Demographics with Attrition Distribution

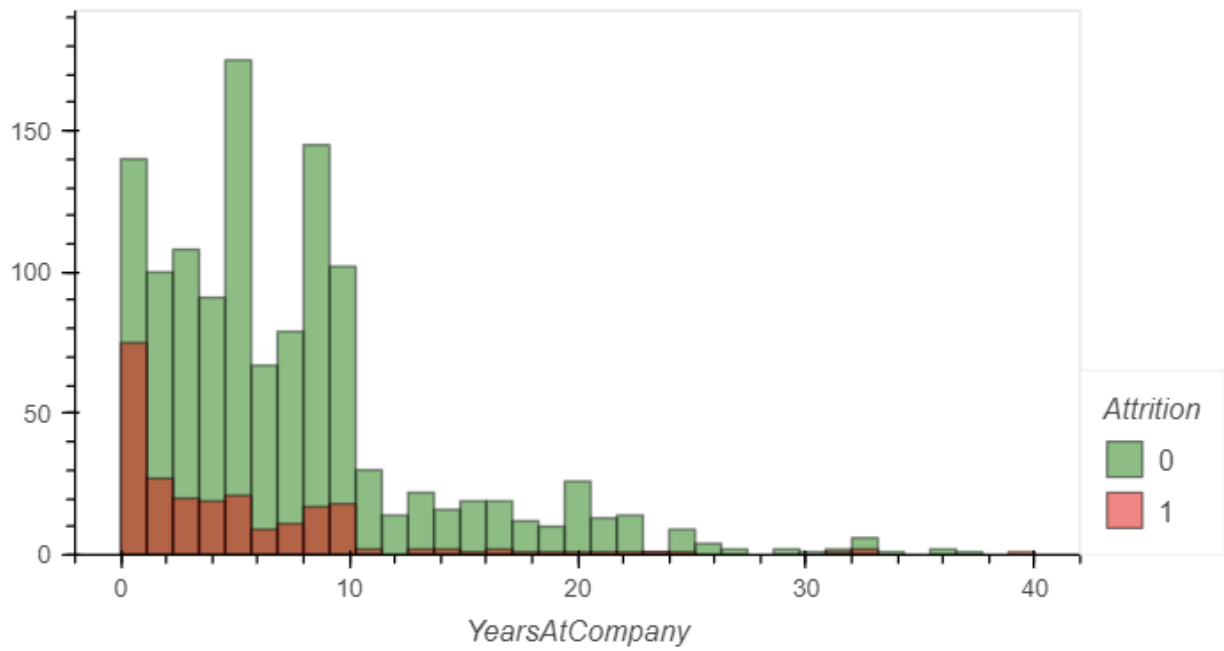


Figure 3 - Number of Years at the Company with Attrition Distribution

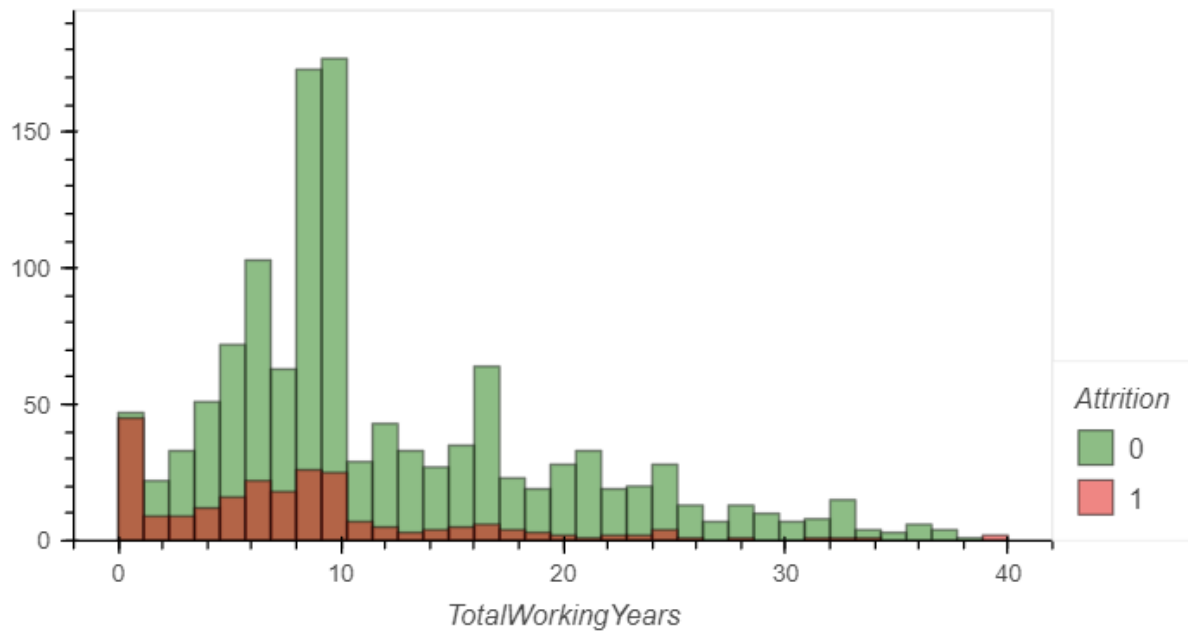


Figure 4 - Total Working Years with Attrition Distribution

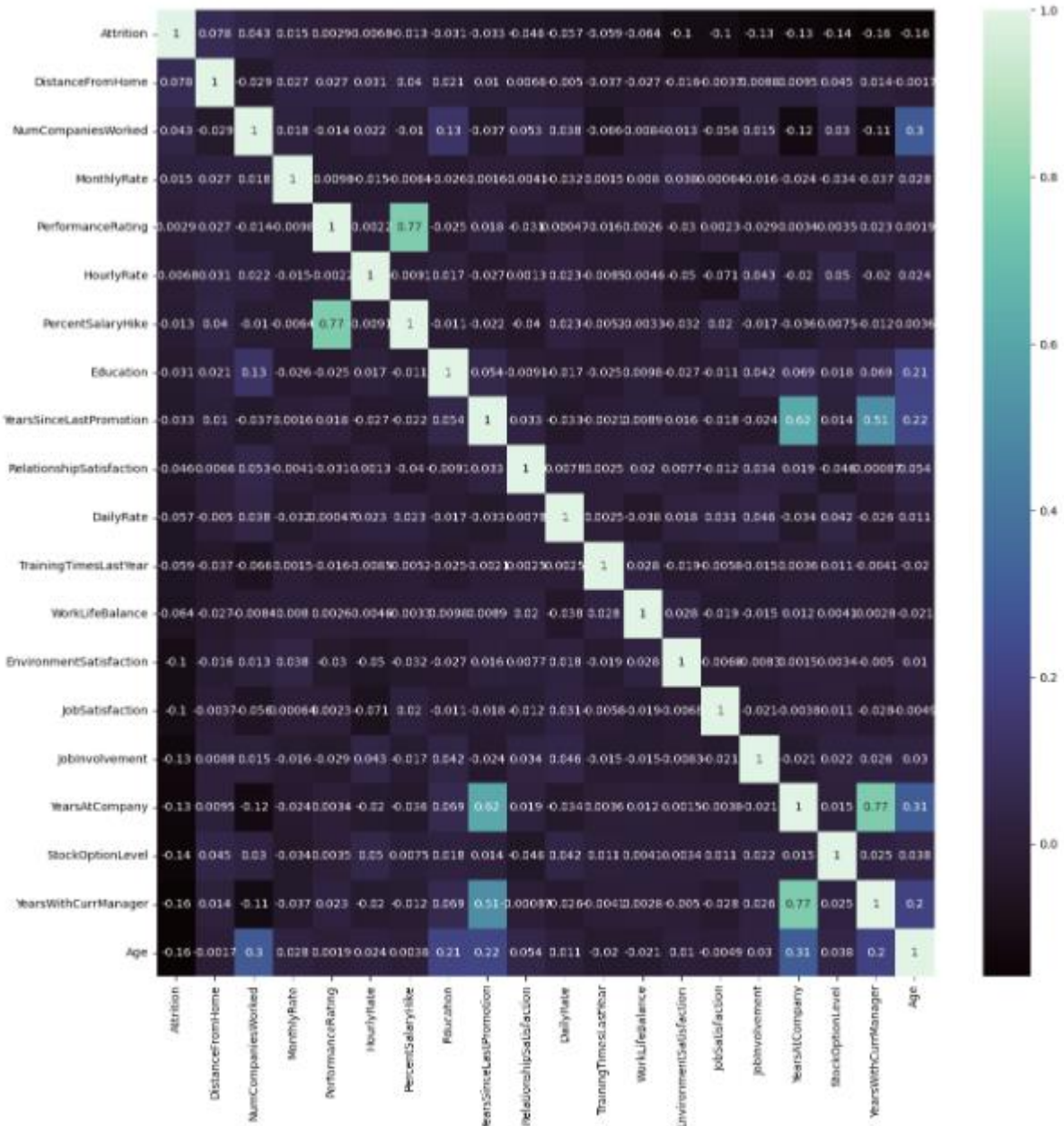


Figure 5 - Correlation Matrix

```
Model: LogisticRegression(penalty='l1', solver='liblinear')
=====
TRAINING RESULTS:
=====
CONFUSION MATRIX:
[[849  14]
 [ 59 107]]
ACCURACY SCORE:
0.9291
CLASSIFICATION REPORT:

```

	0	1	accuracy	macro avg	weighted avg
precision	0.935022	0.884298	0.929057	0.909660	0.926839
recall	0.983778	0.644578	0.929057	0.814178	0.929057
f1-score	0.958780	0.745645	0.929057	0.852212	0.924397
support	863.000000	166.000000	0.929057	1029.000000	1029.000000

```
TESTING RESULTS:
=====
CONFUSION MATRIX:
[[348  22]
 [ 43  28]]
ACCURACY SCORE:
0.8526
CLASSIFICATION REPORT:

```

	0	1	accuracy	macro avg	weighted avg
precision	0.890026	0.560000	0.852608	0.725013	0.836892
recall	0.940541	0.394366	0.852608	0.667453	0.852608
f1-score	0.914586	0.462810	0.852608	0.688698	0.841851
support	370.000000	71.000000	0.852608	441.000000	441.000000

Figure 6 - Logistic Regression Model Results

```
Model: RandomForestClassifier(bootstrap=False, max_depth=10, max_features='auto',
                               n_estimators=500, random_state=42)
=====
TRAINING RESULTS:
=====
CONFUSION MATRIX:
[[863  0]
 [ 15 151]]
ACCURACY SCORE:
0.9854
CLASSIFICATION REPORT:

```

	0	1	accuracy	macro avg	weighted avg
precision	0.982916	1.000000	0.985423	0.991458	0.985672
recall	1.000000	0.909639	0.985423	0.954819	0.985423
f1-score	0.991384	0.952681	0.985423	0.972033	0.985141
support	863.000000	166.000000	0.985423	1029.000000	1029.000000

```
TESTING RESULTS:
=====
CONFUSION MATRIX:
[[360  10]
 [ 63  8]]
ACCURACY SCORE:
0.8345
CLASSIFICATION REPORT:

```

	0	1	accuracy	macro avg	weighted avg
precision	0.851064	0.444444	0.834467	0.647754	0.785599
recall	0.972973	0.112676	0.834467	0.542825	0.834467
f1-score	0.907945	0.179775	0.834467	0.543860	0.790711
support	370.000000	71.000000	0.834467	441.000000	441.000000

Figure 7 - Random Forest Classifier Results

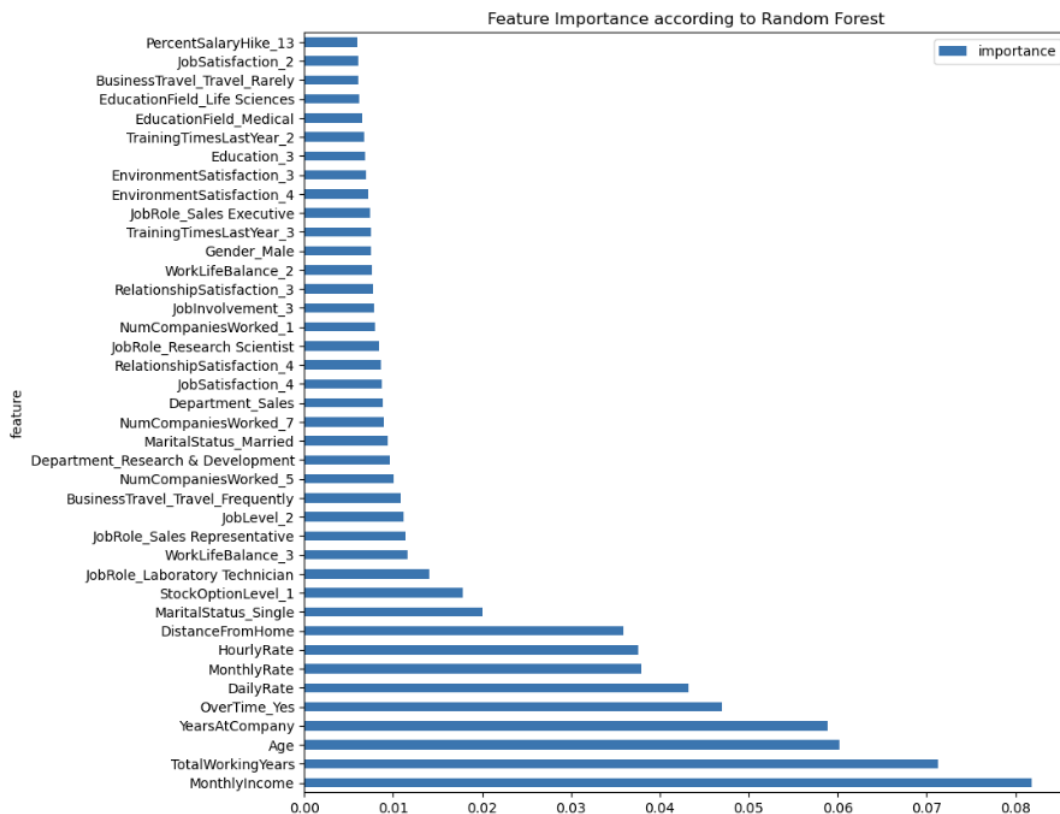


Figure 8 - RF Feature Importance

```

Model: SVC(C=100, gamma=0.001)
=====
TRAINIG RESULTS:
=====
CONFUSION MATRIX:
[[862  1]
 [ 6 160]]
ACCURACY SCORE:
0.9932
CLASSIFICATION REPORT:
              0              1  accuracy  macro avg  weighted avg
precision    0.993088    0.993789    0.993197    0.993438    0.993201
recall       0.998841    0.963855    0.993197    0.981348    0.993197
f1-score     0.995956    0.978593    0.993197    0.987275    0.993155
support     863.000000   166.000000    0.993197   1029.000000   1029.000000
TESTING RESULTS:
=====
CONFUSION MATRIX:
[[346  24]
 [ 42  29]]
ACCURACY SCORE:
0.8503
CLASSIFICATION REPORT:
              0              1  accuracy  macro avg  weighted avg
precision    0.891753    0.547170    0.85034    0.719461    0.836276
recall       0.935135    0.408451    0.85034    0.671793    0.850340
f1-score     0.912929    0.467742    0.85034    0.690335    0.841255
support     370.000000    71.000000    0.85034    441.000000    441.000000

```

Figure 9 - Support Vector Machine Results

RANDOM FOREST	roc_auc_score: 0.543
LOGISTIC REGRESSION	roc_auc_score: 0.546
SUPPORT VECTOR MACHINE	roc_auc_score: 0.500

Figure 10 - ROC AUC Score Comparison

10 Questions:

1. What factor stands out as the most correlated with employee attrition?

Monthly Income (salary)

2. What would you do to help retain employees?

Develop a more competitive salary package, provide increased incentives to stay

3. What other data would be helpful?

Environmental factors, such as pollution or crime rate around the work building

4. Do you think survey results are a good data source?

I think they can be, but I understand that some employees will not respond factually because they do not think the survey is anonymous.

5. What other data source would be helpful?

Based on the results of this analysis, the pay scales of multiple companies would be helpful to prove that is the deciding factor.

6. Is it ethical to study your employees?

It can be as long as you are open and honest about what the study will be used for.

7. Could management be the issue (Marcus Buckingham – “People leave managers, not companies”)?

Yes. From personal experience, I have wanted to quit a job because of management even though I thoroughly enjoyed my work.

8. If management is the issue, what steps can be taken?

Require management to participate in continuing education resources in order to improve their managing skills/style. If the problems continue, be less afraid to make changes at the management level.

9. How would you convince the c-suite that the cost of training new employees outweighs the cost of raises to retain employees?

Just need the numbers. This includes cost of training, time training, loss in productivity, etc.

10. Could environmental factors be an issue?

The short answer is that they could be. They are unlikely to be as big of a deciding factor as salary, but they could still be a push in the direction of leaving.